



10106 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC1275 Filaments MRS				
	1	NGC1275-NORTH-FILAMENT-ALMA	MIRI Medium Resolution Spectroscopy	(8) NGC1275-NORTH_FILAMENT_ALMA
	2	NGC1275-HORSESHOE	MIRI Medium Resolution Spectroscopy	(5) NGC1275-HORSESHOE
	3	NGC1275-SOUTH	MIRI Medium Resolution Spectroscopy	(2) NGC1275-SOUTH
	4	NGC1275-OFF	MIRI Medium Resolution Spectroscopy	(6) NGC1275-OFF
	5	NGC1275-BLUE-LOOP1	MIRI Medium Resolution Spectroscopy	(3) NGC1275-BLUE-LOOP1
	6	NGC1275-BLUE-LOOP2	MIRI Medium Resolution Spectroscopy	(9) NGC1275-BLUE-LOOP2
	7	NGC1275-EAST	MIRI Medium Resolution Spectroscopy	(1) NGC1275-EAST
NGC 1275 MIRI Imaging				
	8	MIRI Imaging Mosaic NORTH FILAMENTS	MIRI Imaging	(7) NGC1275-NORTH

ABSTRACT

We propose MIRI observations of the spectacular 80 kpc-long filamentary nebula around the iconic Perseus cluster central galaxy NGC 1275. This fascinating source is a template for galaxy growth in hot cooling atmospheres in the presence of AGN feedback. The filaments are rich in molecular (CO) gas but form few stars, hence the goals of this proposal: how does multiphase gas cool to form filaments, and why is star formation so inefficient? We will use the MRS and MIRI imager to measure the cooling rate from warm to cold gas (H lines) and the SFR (PAH emission), the missing pieces of the filament puzzle already probed by many facilities. This gas cooling reveals the tightly coupled nature of feedback, and its fate is key to understanding how feedback regulates cooling and star formation. Targeted MRS spectroscopy will provide a complete view of the H

rotational excitation and kinematics, crucial to estimate the molecular hydrogen (H₂) contribution to broad-band filters. Simultaneous large-scale mapping in three MIRI bands will trace gas and dust emission at 50 pc resolution (size of giant CO complexes). The powerful combination of the two will relate the H₂ cooling rate to cold gas and star formation. NGC 1275 is a unique source where MIRI's spatial resolution can link these three complementary viewpoints on cooling flows around brightest cluster galaxies (BCGs) and will provide a reference dataset, since filaments are ubiquitous around all cooling-flow cluster cores (20 to 30% of all clusters). It will challenge our understanding of multiphase gas formation and how star formation is regulated and quenched in one of the most extreme environments of the local Universe.

OBSERVING DESCRIPTION

We perform MIRI imaging and MRS spectroscopy of the filamentary nebula around the Perseus cluster central galaxy: NGC 1275.

MIRI MRS: we target 6 positions in the giant filamentary structure around the iconic central galaxy of the Perseus cluster, NGC1275. 3 of those positions have Spitzer spectra but only 2 lines of H₂ were detected, which prevents a robust estimate of the gas cooling rate. Moreover, no spatial nor kinematical information was available. We request a A+B+C setup which will give a complete 5-27 microns spectral coverage of the H₂ pure-rotational lines (except the 0-0S(0) line), PAH features, and forbidden lines at a spectral resolution of 80-240 km/s. A dedicated sky background (OFF) is common to, and linked to the ON observations in the middle of a non-interruptible sequence. Since the number of groups in FASTR1 would be 100, we use the SLOWR1 readout mode to be compliant with data rates. We use the 4-pt extended source dither pattern which improves PSF sampling and mitigates the effects of cosmic ray showers on the detector.

Since the emission within the MRS FoV will be extended, as recommended, we use a single, 2-pts dithered, OFF background pointing, which will also serve to take simultaneous imaging of the Southern part of the nebula. The OFF position is to the East of the nebula, in a region free of source in WISE, CFHT and 2MASS images.

Simultaneous imaging: We perform simultaneous imaging of the southern nebula in the F560W, F770W and F1000W filters while performing the 6 MRS pointings.

Dedicated imaging: we use a 4-pt extended imager dither to map the Northern part of the filaments which will complete the full mapping of the entire nebula.

We restrict the PA orientation of the MRS observations, to be compliant with the MSZ, PA=68-105, and to allow us to simultaneously image the southern part of the filamentary structure around NGC 1275

Proposal 10106 - Targets - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC1275-EAST	RA: 03 19 50.0500 (49.9585417d) Dec: +41 30 47.00 (41.51306d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES		
	(2)	NGC1275-SOUTH	RA: 03 19 45.4755 (49.9394813d) Dec: +41 29 48.97 (41.49694d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES		
	(3)	NGC1275-BLUE-LOOP1	RA: 03 19 50.8795 (49.9619979d) Dec: +41 30 2.39 (41.50066d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES		
	(5)	NGC1275-HORSESHOE	RA: 03 19 44.9843 (49.9374346d) Dec: +41 31 33.97 (41.52610d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES		
	(6)	NGC1275-OFF	RA: 03 19 58.2168 (49.9925700d) Dec: +41 31 24.36 (41.52343d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES		
	(7)	NGC1275-NORTH	RA: 03 19 45.8527 (49.9410529d) Dec: +41 32 3.65 (41.53435d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Seyfert galaxies] Extended=YES	Proper Motion RA: 9.971237766405153E-6 sec of time/yr Proper Motion Dec: -1.9799995243374724E-4 arcsec/yr Epoch of Position: 2015.5	
	(8)	NGC1275-NORTH_FILAMENT_ALMA	RA: 03 19 48.1760 (49.9507333d) Dec: +41 31 50.20 (41.53061d) Equinox: J2000 Comments: Category=Clusters of Galaxies Description=[Brightest cluster galaxies]		

Proposal 10106 - Targets - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

(9)	NGC1275-BLUE-LOOP2	RA: 03 19 52.7000 (49.9695833d) Dec: +41 30 4.50 (41.50125d) Equinox: J2000
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Comments: This object was generated by the targetselector and retrieved from the NED database.

Category=Clusters of Galaxies

Description=[Perseus Cluster]

Extended=YES

Proposal 10106 - Observation 1 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 1: NGC1275-NORTH-FILAMENT-ALMA												Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NGC1275-NORTH-FILAMENT-ALMA (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(8)	NGC1275-NORTH_FILAMENT_ALMA	RA: 03 19 48.1760 (49.9507333d) Dec: +41 31 50.20 (41.53061d) Equinox: J2000										
	Comments: Category=Clusters of Galaxies Description=[Brightest cluster galaxies]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS			YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	1	SHORT(A)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	1	SHORT(A)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2		IMAGER	F770W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3		IMAGER	F1000W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3	LONG(C)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3	LONG(C)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	

Proposal 10106 - Observation 1 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Aperture PA Range 68.0 to 105.0 Degrees (V3 68.0 to 105.0) Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 2 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 2: NGC1275-HORSESHOE												Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC1275-OFF (Obs 4)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NGC1275-HORSESHOE (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(5)	NGC1275-HORSESHOE	RA: 03 19 44.9843 (49.9374346d) Dec: +41 31 33.97 (41.52610d) Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the NED database.												
	Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	1	SHORT(A)	MRSLONG		SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	1	SHORT(A)	MRSSHORT		SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	2		IMAGER	F770W	SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	3		IMAGER	F1000W	SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	3	LONG(C)	MRSLONG		SLOWR1	12	3	1	Dither 1	4	12	3631.268	
	3	LONG(C)	MRSSHORT		SLOWR1	12	3	1	Dither 1	4	12	3631.268	

Proposal 10106 - Observation 2 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 3 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 3: NGC1275-SOUTH												Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC1275-OFF (Obs 4)]												
	Comments: Position 2 of IRS from Johnstone et al.												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NGC1275-SOUTH (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC1275-SOUTH	RA: 03 19 45.4755 (49.9394813d) Dec: +41 29 48.97 (41.49694d) Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the NED database.												
	Category=Clusters of Galaxies												
	Description=[Perseus Cluster] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	1	SHORT(A)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	1	SHORT(A)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2		IMAGER	F770W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3		IMAGER	F1000W	SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3	LONG(C)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992	
	3	LONG(C)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992	

Proposal 10106 - Observation 3 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 4 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 4: NGC1275-OFF												Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [NGC1275-HORSESHOE (Obs 2), NGC1275-SOUTH (Obs 3), NGC1275-BLUE-LOOP1 (Obs 5), NGC1275-BLUE-LOOP2 (Obs 6), NGC1275-EAST (Obs 7)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NGC1275-OFF (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(6)	NGC1275-OFF	RA: 03 19 58.2168 (49.9925700d) Dec: +41 31 24.36 (41.52343d) Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the NED database.												
	Category=Clusters of Galaxies Description=[Perseus Cluster] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F560W	All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	1	1	Dither 1	2	2	573.358	
	1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	2	2	573.358	
	1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	2	2	573.358	
	2		IMAGER	F770W	SLOWR1	12	1	1	Dither 1	2	2	573.358	
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	2	2	573.358	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	2	2	573.358	
	3		IMAGER	F1000W	SLOWR1	12	1	1	Dither 1	2	2	573.358	
	3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	2	2	573.358	
	3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	2	2	573.358	

Proposal 10106 - Observation 4 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 5 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 5: NGC1275-BLUE-LOOP1												Fri Mar 13 18:05:23 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[NGC1275-OFF (Obs 4)]													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(NGC1275-BLUE-LOOP1 (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	NGC1275-BLUE-LOOP1	RA: 03 19 50.8795 (49.9619979d) Dec: +41 30 2.39 (41.50066d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the NED database.													
	Category=Clusters of Galaxies													
	Description=[Perseus Cluster] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	1	SHORT(A)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	1	SHORT(A)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2		IMAGER	F770W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3		IMAGER	F1000W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3	LONG(C)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3	LONG(C)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		

Proposal 10106 - Observation 5 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 6 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 6: NGC1275-BLUE-LOOP2												Fri Mar 13 18:05:23 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[NGC1275-OFF (Obs 4)]													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(NGC1275-BLUE-LOOP2 (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(9)	NGC1275-BLUE-LOOP2	RA: 03 19 52.7000 (49.9695833d) Dec: +41 30 4.50 (41.50125d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the NED database.													
	Category=Clusters of Galaxies													
	Description=[Perseus Cluster] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	1	SHORT(A)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	1	SHORT(A)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2		IMAGER	F770W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3		IMAGER	F1000W	SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3	LONG(C)	MRSLONG		SLOWR1	12	2	1	Dither 1	4	8	2388.992		
	3	LONG(C)	MRSSHORT		SLOWR1	12	2	1	Dither 1	4	8	2388.992		

Proposal 10106 - Observation 6 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 7 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 7: NGC1275-EAST												Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC1275-OFF (Obs 4)]												
Comments: East Position of the Spitzer IRS SH Slit of Johnstone et al.													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NGC1275-EAST (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC1275-EAST	RA: 03 19 50.0500 (49.9585417d)										
			Dec: +41 30 47.00 (41.51306d)										
			Equinox: J2000										
Comments: This object was generated by the targetselector and retrieved from the NED database.													
Category=Clusters of Galaxies													
Description=[Perseus Cluster]													
Extended=YES													
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	2		IMAGER	F770W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	3		IMAGER	F1000W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
	3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	

Proposal 10106 - Observation 7 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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Proposal 10106 - Observation 8 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Observation	Proposal 10106, Observation 8: MIRI Imaging Mosaic NORTH FILAMENTS										Fri Mar 13 18:05:23 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 8:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(7)	NGC1275-NORTH	RA: 03 19 45.8527 (49.9410529d) Dec: +41 32 3.65 (41.53435d) Equinox: J2000				Proper Motion RA: 9.971237766405153E-6 sec of time/yr Proper Motion Dec: -1.9799995243374724E-4 arcsec/yr Epoch of Position: 2015.5				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Galaxy										
	Description=[Seyfert galaxies]										
	Extended=YES										
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	2	10.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	4	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	50	1	1	Dither 1	16	16	2220.032	
	2	F770W	FASTR1	50	1	1	Dither 1	16	16	2220.032	
	3	F1000W	FASTR1	50	1	1	Dither 1	16	16	2220.032	
	4	F1800W	FASTR1	50	1	1	Dither 1	16	16	2220.032	

Proposal 10106 - Observation 8 - Gas cooling and star formation in the Perseus cluster filaments around NGC 1275

Special Requirements	Group Visits within 53.0 Days Visits Same PA Same Aperture PA 1, 2, 3, 4, 5, 6, 7, 8 (V3 PAs differ)
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